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Muscle Pain Code: a novel tool for screening the risk of time-loss muscle injury in professional male football players

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Muscle Pain Code: a novel tool for screening the risk of time-loss muscle injury in professional male football players

ABSTRACT

Objective: To investigate the association of a novel post-match muscle pain map, named Muscle Pain Code (MPC), with the risk of subsequent time-loss muscle injury in a cohort of professional football (soccer) players.

Methods*:* The MPC classifies pain in four codes: code 0, "*no pain*"; code 1, "*generalized muscle pain*"; code 2, "*diffused site muscle pain*"; and code 3, "*specific site muscle pain*". Over four consecutive seasons, MPC was collected on the second post-match day and players were followed for occurrence of time-loss muscle injury over the next five days. Players exposed to at least 45 minutes in two consecutive matches within seven days were included as cases for analysis.

Results*:* Eighty players participated in the study. Of 1,656 cases analyzed, 229 resulted in time-loss muscle injuries. Only 2% of cases with codes 0 and 1 resulted in time-loss muscle injuries. Conversely, 63% and 78% of codes 2 and 3 were followed by time-loss muscle injuries, respectively. Compared with the reference scenario (i.e., code 0 on MPC), the risk of subsequent time-loss muscle injury was significantly higher when players recorded code 2 (odds ratio, 4.29; 95%CI, 3.62 to 4.96) or code 3 (odds ratio, 5.01; 95%CI, 4.05 to 5.98) on MPC, but not when they recorded code 1 (odds ratio=-0.27; 95%CI, 1.05 to 0.56).

Conclusions: Players experiencing well-outlined pain area on the second post-match day were more likely to incur a time-loss muscle injury in the subsequent days compared to those experiencing spreading pain or no pain.

Key words: injury prevention, injury risk, injury prediction; pain measurement, soccer.

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INTRODUCTION

According to the *International Olympic Committee consensus statement* [1], injury is tissue damage or other derangement of normal physical function due to participation in sports, resulting from rapid or repetitive transfer of kinetic energy. The *Munich consensus statement* on the terminology and classification of muscle injuries in sport [2] classifies muscle-related complaints as either "structural muscle injuries" or "functional muscle disorders," depending on the presence of macroscopic muscle tear. Functional disorders can manifest as "overexertion-related muscle disorders," such as delayed-onset muscle soreness, or "neuromuscular disorders" stemming from spine or muscle origins [2]. Moreover, there are “direct muscle injuries” resulting from direct impact at the site, such as contusions [2]. When any of those situations hinders an athlete's to fully participate in training or competition, it is termed a 'time-loss injury', employed as the primary measure for assessing injuries in epidemiological research.

In professional male football (soccer) players, the overall incidence of time-loss injuries is 8.1 injuries/1,000 hours [3]. Muscle injuries account for 44-47% of all time-loss injuries [4] and play a negative role in teams’ performance [5] and clubs’ finances [6]. Consequently, muscle injury prevention is a major concern of the football medicine community. Elite clubs have been investing in preseason screenings to identify those players with a high risk of injury [7,8]. However, considering the complexity of sports injuries [9] and the non-linear dynamic behavior of athletes and their risk factors for injuries along the season [10], preseason data has poor predictive value for in-season injuries [11]. Therefore, incorporating continuous athlete monitoring to identify when players are most susceptible to injury seems recommendable for injury prevention [12].

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To enhance ecological value in professional football settings, efficient strategies with minimal physical burden on players can be utilized throughout the season to inform decisions regarding their exposure to training and matches [13]. Self-reported measures of muscle pain offer affordability, quickness, and ease of application within sports settings. These measures typically utilize ordinal scales to quantify the intensity of painful symptoms, such as the visual analogue scale (VAS) or the numeric rating scale (NRS) [14]. Interestingly, previous evidence suggests that muscle pain measured 2 to 3 days after a match is associated with an increased risk of injury in the following match [15]. It is plausible that delayed recovery kinetics from muscle pain after a match may arise from an overexertion-related muscle disorder, a neuromuscular disorder, or a contusion. In all three instances, as long as the player continues to participate in training and/or matches, this muscle pain is typically reported as a complaint but not classified as a time-loss injury. Nevertheless, it remains unclear whether players experiencing post-match muscle pain attributable to an overexertion-related muscle disorder, a neuromuscular disorder, or a contusion are at a higher risk of a subsequent time-loss injury.

It turns out that similar levels of pain intensity can result from an overexertion-related muscle disorder, a neuromuscular disorder, and a contusion. However, the mapping of pain differs between these conditions: individuals with an overexertion-related muscle disorder usually experience a widespread discomfort across the exercised muscle groups, whereas those with a neuromuscular disorder or a contusion often outline a specific site or zone of muscle pain [2]. Based on this rationale, we have introduced a novel mapping method for assessing muscle pain: the Muscle Pain Code (MPC). The MPC classifies muscle pain into 4 codes: "*no pain*" (code 0), "*generalized*

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muscle pain" (code 1), "*diffused site muscle pain*" (code 2) and "*specific site muscle pain*" (code 3). The MPC was integrated into the post-match monitoring routine of a professional football team for four seasons. The aim of this study was to investigate the correlation of MPC at the second post-match day with the risk of subsequent time-loss muscle injury.

METHODS

Study design

This cohort study analyzed routinely collected data from a professional men's football team throughout four consecutive competitive seasons in the Brazilian calendar, from March to December (i.e., pre-season data were not included in the study). The same set of players and timeframe were analyzed in a prior study assessing the predictive value of CK concentration for the risk of subsequent muscle injuries [16].

As part of the weekly clinical and physiological monitoring routine, MPC was collected on the second day after matches, before returning to the training routine. All players who played for the club during the study were informed and familiarized before having their data included in the analysis. Follow-up for the occurrence of indirect muscle injuries continued for the next five days. A second match was necessarily played into four to seven days after the first match (i.e., the index match) for a case be eligible for analysis in the current study. The study flowchart is presented in Figure 1.

<< FIGURE 1>>

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Participants

All players from a Brazilian male professional football team along the four seasons of the study were considered for eligibility. The team competed in the Brazilian first and second league divisions, Brazil's Cup, CONMEBOL Sudamericana Cup, and state competitions during the study data collection. The number of players that met the inclusion criteria over the study determined the convenience sample size. All players voluntarily signed an informed consent to participate in this study, approved by the Ethics Committee on Human Research of the XXX (# 2.390.889).

To be included, a player should: 1) have gone through a general health assessment routine conducted by healthcare staff and be cleared to compete; 2) have been monitored after playing at least five matches in the club, as a cutoff to avoid biased data; 3) have played at least 45 minutes in the index match without sustaining injuries; and 4) have played at least 45 minutes in the second match (except cases in which he suffered a muscle injury during the second match). Only periods in which index and second matches were interspersed by four to seven days were included in the analysis (i.e., congested periods with three matches played within a 7-day period were not included). Goalkeepers were not included in the analysis.

Muscle Pain Code (MPC) assessment

A categorical 4-point pain scale named Muscle Pain Code (MPC) was designed by authors to players report their post-match perception of the extension of the painful area in (Figure 2): "*no pain*" (coded as 0), "*generalized muscle pain*" (coded as 1), "*diffused site muscle pain*" (coded as 2) and "*specific site muscle pain*" (coded as 3).

<< FIGURE 2 >>

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The MPC was applied by the team's physiologist on the second post-match day. The player was queried regarding any muscle pain in the lower limbs at rest or during routine daily activities, such as walking or ascending and descending stairs. If the player indicated no muscle pain, the evaluator assigned a code of 0 on the MPC. Conversely, if the player experienced pain in one or more muscle groups in the lower limb, they were prompted to assign a corresponding code based on the MPC (Figure 2). If different codes were inferred for different lower limbs regions, the highest code was recorded and utilized for subsequent analysis.

Players with code 0 on MPC were allowed to immediately participate in the team's planned training sessions and matches. Players with code 1 on MPC participated in the scheduled training sessions after attending post-match recovery strategies (e.g., cold-water immersion, manual therapy, electrotherapy, stretching), according to their individual preference. When players' pain perception was recorded as code 2 or code 3 on MPC, the physical therapist guided a clinical evaluation consisting of palpation of the muscles, passive and active range of motion, functional performance, and physical complaint. If a muscle injury was suspected, the team physician conducted an in-depth assessment, deciding whether ultrasound [17] or magnetic resonance images [18] were necessary for the diagnostic. When a muscle injury requiring absence from sporting activities was confirmed, the player was referred for treatment by the club's physical therapy department.

Injury follow-up

Time-loss injury was defined as any physical complaint resulting from football training or match exposure that prevented a player from participating regularly in at

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least one training or match [19]. We included acute indirect muscle injuries with macroscopic evidence of muscle tear (i.e., structural muscle injuries) and with no macroscopic evidence of muscle tear (i.e., functional muscle disorders) [2]. All potential injury cases were assessed by the club's medical doctor and physical therapist. Imaging exams were carried out when needed for diagnostic support. The player's removal from team activities in the following days (i.e., a time-loss injury case for the present study) was taken in a joint decision of the healthcare staff.

Statistical analysis

Descriptive statistics were used to summarize participants' characteristics. Means, standard deviations (SD) and 95% confidence intervals (CI) were computed for continuous variables and frequency distributions were generated for categorical variables. First, the association between MPC and subsequent muscle injury was assessed through univariate logistic regression analysis. Then, generalized estimating equation (GEE) model with an unstructured working correlation was used to evaluate the longitudinal associations of MPC with the risk of muscle injury. Associations were expressed as OR with 95% CI. A $p<0.05$ was considered significant. Analysis was performed using an IBM SPSS Statistics (version 22, Armonk, NY).

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RESULTS

Participants' mean (SD) characteristics were: 24.9 (4.1) years old, 79.0 (6.0) kg in body mass, and 180.0 (5.5) cm in height. During the 4-season period, the team played a total of 273 matches (mean of ~68 per season). Each player generated between 5 and

82 eligible cases for analysis (median of 16 cases), resulting in 1,656 cases included in this study. Muscle injuries occurred in 229 cases (14%), all during official matches. Sixty-six out of the 80 players included in the study sustained at least one muscle injury during the study. The hamstrings were the most injured muscle group (123 cases, 54% of all muscle injuries), followed by the adductor (49 cases), triceps surae (43 cases), quadriceps (11 cases), and psoas (3 cases).

Out of 963 cases with code 0, only 21 (2%) resulted in time-loss muscle injuries. Similarly, only 6 of the 389 cases (2%) cases with code 1 resulted in time-loss muscle injuries. Conversely, 143 out of 230 cases (63%) with code 2 were followed by time-loss muscle injuries, as well as 59 out of 76 cases (78%) with code 3. In overall, codes 2 and 3 on MPC preceded 88% of all time-loss muscle injuries.

According to the univariate logistic regression analysis, MPC was associated with muscle injury with a significant prediction likelihood (b=2.225, SE=0.126, OR=9.258, 95% CI (7.23-11.86), p<0.001) (Figure 3). Compared with the reference scenario (MPC 0, "*no pain*"), the risk of subsequent time-loss muscle injury was 4- to 5-fold increased when players reported codes 3 and 4, respectively (Table 1).

<< FIGURE 3 >>

<< TABLE 1 >>

DISCUSSION

In this 4-season cohort study, we investigated the association between a novel method to assess post-match muscle pain, named Muscle Pain Code (MPC), with subsequent time-loss muscle injury occurrence in a cohort of professional football players. To the best of our knowledge, this is the first study to propose a classification

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of muscle pain based on the extent of the pain area and follow-up for the incidence of muscle injury in subsequent days. Our main finding was that players experiencing well-outlined pain area on the second post-match day were more likely to incur a time-loss muscle injury in the subsequent days compared to those experiencing spreading pain or no pain.

In the current study, time-loss muscle injuries occurred in 14% of cases, represented by players exposition to at least 45 minutes of play in two consecutive matches within seven days. The hamstring, adductors, gastrocnemius, and quadriceps were the most prevalent muscles injured, in agreement with football epidemiological studies [3, 20]. However, the average number of ~57 muscle injuries per season in the current study is higher than previously reported in a single-season study with South American teams (~20 muscle injuries per team) [4]. Although concerning, this inflated incidence of muscle injuries may partially be explained by the definition of injury used in the current study, based on the absence of a player from training or match due to muscle-related complaints [19]. This means that some players were spared from full participation in the team's regular routine due to structural muscle injuries and other due to functional muscle disorders by a joint decision made by the healthcare staff aiming to minimize symptoms that could potentially pose risks or harm to the players' participation.

The MPC development was based on the rationale that individuals with an overexertion-related muscle disorder typically experience widespread discomfort across exercised muscle groups, while those with a neuromuscular disorder or a contusion often pinpoint specific sites or zones of muscle pain [2]. However, the methodological design of the current study does not allow us to state that the higher extended area of

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pain characterized in code 1 of the MPC resulted exclusively from an overexertion-related muscle disorder. Similarly, the more localized pain reported by players who scored codes 2 and 3 on the MPC cannot be solely attributed to a neuromuscular disorder or a contusion. Thus, the information provided by the current study is more directed towards practical implications rather than mechanistic insights regarding the MPC.

Studies have repeatedly emphasized the injury history as the most influential risk factor for time-loss muscle injuries in athletes [21-23]. Conversely, the impact of modifiable risk factors continues to produce conflicting findings [21-23]. In this context, the 4 to 5-fold increase in the risk of subsequent time-loss muscle injury in players with localized muscle pain is an impactful finding of the present study. It is noteworthy that a history of hamstring strain injury in the current season leads to a risk ratio of 4.8 (95%CI 3.5 to 6.6) [22]. This suggests that the athlete's likelihood of sustaining a time-loss muscle injury with a code 2 or 3 on the MPC on the second post-match day seems comparable to a recent muscle injury.

In a 2-season cohort study, Delaval et al. [15] investigated the relationship between the post-match recovery kinetics and the occurrence of any kind of non-contact injuries in a professional football club, including a Hooper scale for describing muscle soreness (from 1 to 7) collected up to three days after matches. Based on that pain intensity scale, the authors reported an association between the level of pain reported by players and the risk of injuries in subsequent days. When pain level was assessed two days after the match, Delaval et al. [15] found a relative risk of 1.48 (95%CI 1.03 to 2.13). This relative risk was increased to 2.08 (95%CI 1.86 to 2.32) when the pain assessment was performed on the third post-match day. Interestingly, those authors

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reported a lack of correlation between biochemical analysis of creatine kinase concentration and the risk of subsequent injuries [15], in alignment with observations from a preceding study that collected data concurrently with the present study [16]. Taking together, ours and Delaval's findings encourage healthcare staff to include pain monitoring in the football teams' routines.

Some limitations of this study should be acknowledged. First, the specific muscle groups associated with each MPC code reported by the player were not documented in the healthcare staff records. As a result, data analysis accounted for the highest MPC value on each post-match screening and the occurrence or absence of subsequent time-loss muscle injury, regardless of whether the injured muscle corresponded to the one for which the player reported the specific code. Second, we did not have access to how many of the registered time-loss muscle injuries were diagnosed as structural muscle injuries or functional muscle disorders. While nearly all structural muscle injuries necessitate the player's withdrawal from their sporting activities, the management of functional muscle disorders is entirely contingent upon the clinical assessment and is affected by several factors analyzed by the healthcare staff (i.e., time of the season, coach need/preference, concern of injury due to other intrinsic factors such as injury history, age, etc.). Third, a segment of data regarding the duration of players' absence due to time-loss muscle injuries was lost due to changes within the club's healthcare staff. Consequently, it was not feasible to ascertain whether the MPC correlates with the duration of time away. Fourth, match demands data (i.e., internal and external workload measures) and post-match individual issues (e.g., sleep quantity and quality, diet, hydration, specific recovery strategies) are potential confounders not addressed in this study.

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The aforementioned limitations should not invalidate the findings of the current study. To advance the understanding of the predictability of the MPC for time-loss muscle injuries, we encourage future studies to provide more detail on the profile of time-loss muscle injuries and incorporate the recording of the muscle group together with the MPC codes. During our data collection, we noticed that some athletes may find it difficult to distinguish between code 2 and 3 depending on the size of the painful area. Nevertheless, considering the comparable odds ratios of these codes in our study, the practical implication remains consistent in such scenarios. Finally, we emphasize that we do not propose replacing the already known pain intensity scales with the MPC. On the contrary, it is conceivable that the extent of the area and the intensity of pain could be used synergistically to increase the decision-making capacity of the healthcare staff.

CONCLUSION

Professional male football players who recorded codes 2 or 3 on the second post-match day experienced a 4- to 5-fold increased risk of sustaining a muscle injury in the subsequent days. Further investigation is warranted to clarify the predictive value of this instrument. Nonetheless, the MPC emerges as an affordable, quick, and easy-to-apply measure to integrate into the screening protocol for assessing the risk of muscle injury in the 'real world' of professional football.

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TABLES

Table 1. Longitudinal analysis of Muscle Pain Code (MPC) with the risk of time-loss muscle injury in professional football players.

MPC	OR	SE	(95%CI)	p
0	reference	reference	reference	reference
1	-0.27	0.41	-1.05 – 0.56	0.549
2	4.29	0.34	3.62 – 4.96	<0.001*
3	5.01	0.49	4.05 – 5.98	<0.001*

**Statistically significant difference (p-value<0.05); CI, confidence interval; OR, odds ratio; SE, standard error. Code zero was used as reference value.*

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FIGURE LEGENDS

Figure 1. Flowchart of the study.

Figure 2. Illustrative diagram of the Muscle Pain Code (MPC). Painful areas are represented in red.

Figure 3. Likelihood of time-loss muscle injury in the five subsequent days (0 = not injured; 1 = injured) for each Muscle Pain Code (MPC) collected two days after the index match.

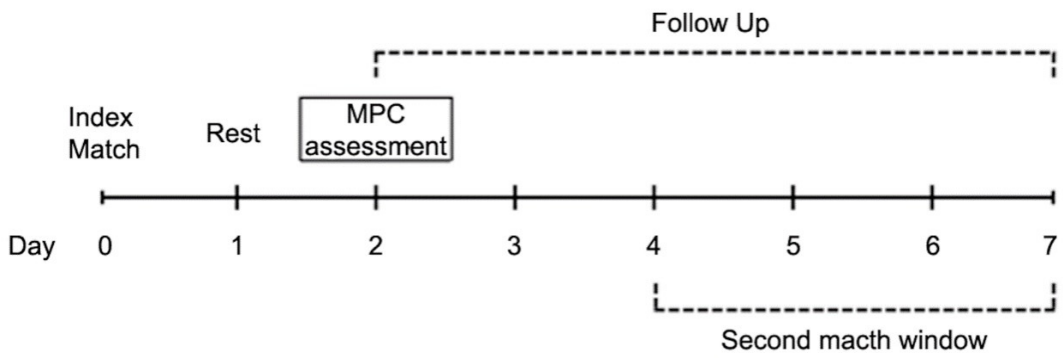

FIGURE 1

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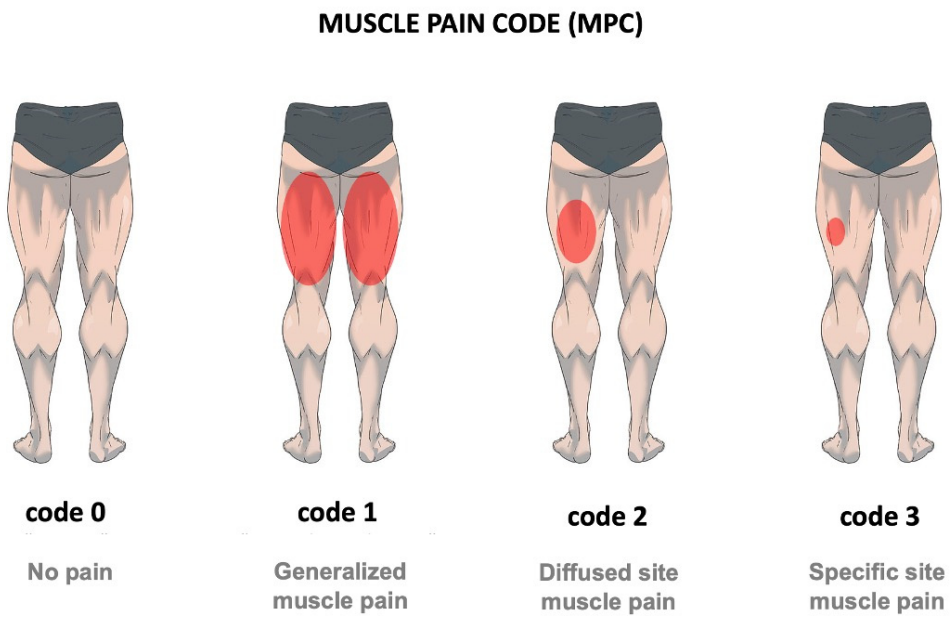

FIGURE 2

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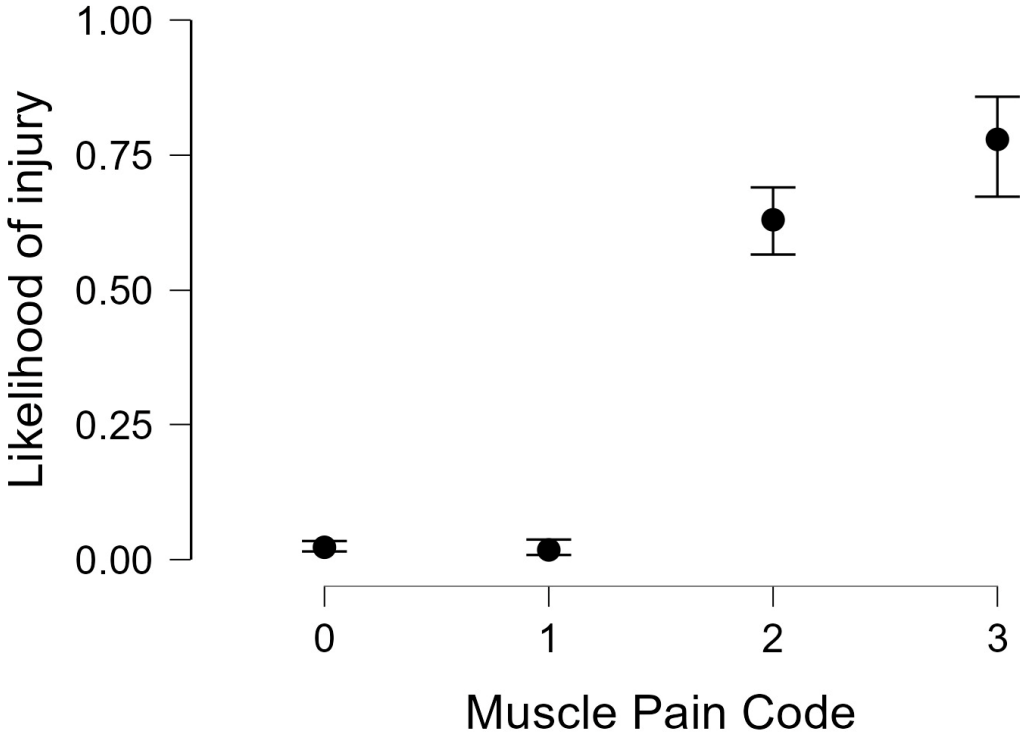

FIGURE 3

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